GeoBalance: Agentic Design of a Deep-Geothermal Heating and Cooling System

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Abstract

A district needs 30 MW of heating and 10 MW of cooling from one geothermal reservoir whose brine carries methane and CO_2. We built an open-weight LLM agent that designs the system by calling the GEMINI digital twin and our own hand-calculation models, runs a 365-day simulation, and improves the design until a reviewer in code accepts it. GEMINI shows that the electric submersible pump, not the reservoir, limits each doublet to about 235–249 m³/h. Going from 2,500 m to 3,400 m raises the brine temperature from 82 °C to 109 °C at the plant, so the network can be heated directly without a heat pump. The recommended design, three doublets at 3,400 m with direct heating, an electric chiller and a gas engine on the associated methane, meets all demand with no backup boiler and emits 5,533 t CO_2 per year, 79% below a gas-boiler reference; reinjecting the gas would reach 94%. Our hand-calculation tools are verified by 12 automated tests and cross-checked against GEMINI.

1 Introduction

The GEMINI Hackathon challenge [1] asks for an integrated system that supplies a district with heating at 100 °C and cooling at 20 °C from a single geothermal reservoir. The two demand peaks fall in opposite seasons (Table 1), and the brine releases gas that must be handled. The task also requires an agentic AI workflow that proposes, simulates, checks and improves designs.

Our objective was a design with zero limit violations, a backup boiler below 5% of annual heat, and the lowest possible CO_2 emissions, with every number traceable to a model rather than to the language model itself. We use the GEMINI digital twin [2] for wells and pumps.

Table 1. Challenge inputs and constraints [1].

Item	Value
Heating demand	30 MW peak (28 January), 111.67 GWh/yr, supply 100 °C
Cooling demand	10 MW peak (19 July), 23.74 GWh/yr, supply 20 °C
Reservoir	100 m thick, 200 mD, 250 bar at 2,500 m, gradient 30 °C/km from 12 °C; deeper wells allowed
Brine and gas	1,030 kg/m³, 0.4 mPa·s; gas–liquid ratio 0.8 Nm³/m³ (80% CH_4, 20% CO_2)
Constraints	1–3 doublets, 50–300 m³/h each, brine cooling ≤ 40 K, 330 operating days per doublet, grid 0.3 kg CO_2/kWh

2 Data and methods

2.1 Subsurface model and hand calculations

Reservoir temperature follows the given gradient, and the heat delivered by one doublet follows from the brine flow and its temperature drop:

$$T = 12\ °C + 30\ °C/km \cdot z \qquad Q = (q/3600) \cdot \rho \cdot c_p \cdot \Delta T \tag{1, 2}$$

with c_p = 4,000 J/(kg·K) (our assumption). The pressure difference of a steady-state doublet with Darcy flow and no skin is

$$\Delta p = q \cdot \mu \cdot ln(L / r_w) / (\pi \cdot k \cdot h) \tag{3}$$

with well spacing L = 1,500 m and well radius r_w = 0.1 m (our assumptions). These relations, together with models for the heat pump (COP 3), absorption chiller (COP 0.7, drive ≥ 80 °C), electric chiller (COP 5) and gas engine (40% electric efficiency), are implemented as tools in `tools.py`. Every tool returns a list of warnings when a limit is broken instead of silently capping the value. Pressure limits are set at injector ≤ +25–50% and producer ≥ 75% of reservoir pressure.

2.2 GEMINI integration and well sizing

Our wrapper builds a GEMINI simulator project from the design variables, runs the producer and injector nodal solves, and returns results plus warnings. We set the official values instead of the example defaults:

- **Reservoir:** 250 bar at 2,500 m (hydrostatic at other depths), temperature 12 °C + 30 °C/km, productivity index 14.4 m^3/h/bar (our estimate from 200 mD, 100 m, 0.4 mPa·s).
- **Gas:** 0.8 Nm^3/m^3, density 0.969 kg/Nm^3, 29.6 MJ/kg, 68.4 kg CO_2/GJ (includes the CO_2 already in the gas). All gas goes to the GEMINI CHP for the cross-check.
- **Heat pump:** GEMINI's Carnot formula (efficiency 0.5) with the mean source temperature, called directly because the model is not in the simulator registry.
- **Plant temperature:** GEMINI gives the brine temperature at the plant, about 5 K below the reservoir because of wellbore, separator and filter losses.

2.3 Annual simulation and design variables

Each design is evaluated over 365 daily steps against the demand profiles. Design variables are depth, number of doublets, flow per doublet, heating route (direct, preheat plus heat pump, or heat pump only), cooling route (absorption or electric chiller), gas use (engine, flare or reinjection) and the start day of each 35-day maintenance outage. Our assumptions: network return temperature 50 °C, heat-exchanger approach 5 K, backup gas boiler efficiency 92% (0.202 kg CO_2/kWh fuel) and grid 0.3 kg CO_2/kWh. The simulation reports energy balances, grid electricity, CO_2, equipment sizes, violations and warnings. The CO_2 reference is our assumption: all heat from a gas boiler and all cooling from an electric chiller, 25,943 t CO_2/yr.

2.4 Agentic AI workflow

Our agent is an open-weight LLM that designs the system by calling engineering tools, while a code-based reviewer checks its work before it may answer. The LLM never computes a result itself: every number comes from GEMINI or our surface models (Figure 1).

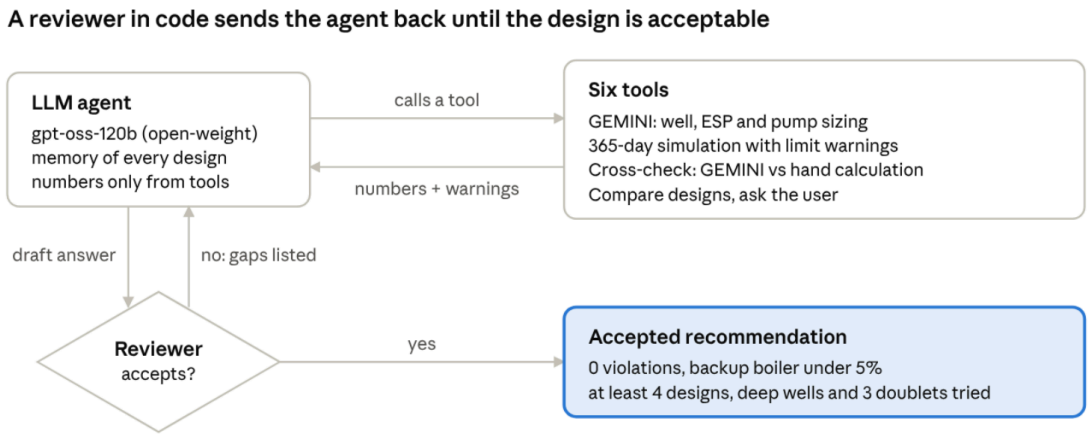


Figure 1. GeoBalance agent architecture: 6 tools, 1 reviewer gate. The agent calls tools for every number; the reviewer sends it back until its work is complete and acceptable.

- **LLM:** gpt-oss-120b [3] (open-weight, released by OpenAI), served through Groq's free OpenAI-compatible API. The short evidence tests used gpt-oss-20b, because the 120b daily free limit (200,000 tokens) ran out. Any OpenAI-compatible endpoint works, including a local Ollama server.
- **Tools (6):** get_challenge_inputs (fixed inputs and demand summary), size_doublet (GEMINI well and ESP sizing), evaluate_design (365-day simulation), cross_check_hand_calc (GEMINI vs. independent hand calculation), compare_designs (table from memory), ask_user (missing or contradictory input).
- **Reasoning:** the system prompt sets the rules: numbers only from tools, fix one variable at a time, compare shallow vs. deep, 2 vs. 3 doublets, heat pump vs. direct heating, and the gas options.
- **Memory:** every evaluated design is stored in memory.json. A one-line summary per design goes into the next prompt, so the agent does not repeat runs.
- **Reviewer (critic step):** before a final answer is accepted, code checks it. It needs at least 4 designs, a deep well, 3 doublets and direct heating, and a recommendation that is acceptable (no violations, backup boiler under 5%). Otherwise the agent is sent back with the gaps, up to 3 times. It only reviews design work done in the same session.
- **Physics guard rails:** tools return violations and warnings instead of crashing. GEMINI results with 0 bar pump intake are rejected as non-physical. ESP intake must stay above 35 bar (free gas), flow within 50–

300 m³/h, brine cooling at most 40 K. Two doublets offline at once is a violation; an outage in the heating season is a warning.

- **Token saving and robustness:** compact tool outputs; older results are trimmed to their verdict (a test conversation shrank from 4,909 to 2,218 tokens). Rate limits (413/429) trigger trim, wait and retry; API-rejected tool calls are fed back to the agent; the transcript is always saved.

2.5 Validation approach

The tools were checked in two ways: (i) 12 automated tests against results known to be correct, and (ii) a cross-check of GEMINI against the independent hand calculations for the same operating point.

3 Results

3.1 Depth and well capacity

At the given depth of 2,500 m the brine is too cold for the 100 °C supply; at 3,400 m it can heat the network directly (Table 2). The hand calculation (Eq. 3) shows that 300 m³/h needs only about 52 bar of total pressure difference, well inside the 125–190 bar allowed by the pressure limits, so the reservoir is not the constraint. With the official reservoir inputs, the pump and tubing limit each doublet to about 235–249 m³/h, not the 300 m³/h cap. Above that, the ESP intake falls below 35 bar and free gas would reach the pump (Table 3).

Table 2. Effect of depth.

Depth	Reservoir temp.	Brine at plant	Pressure	Consequence
2,500 m (given)	87 °C	82 °C	250 bar	Below 100 °C supply: heat-pump top-up needed
3,400 m (chosen)	**114 °C**	**109 °C**	**343.5 bar**	**Direct heating to 100 °C, no heat pump**

Table 3. Maximum acceptable flow per doublet from GEMINI (ESP intake ≥ 35 bar).

Depth	Brine at plant	Max flow	ESP setup	ESP power	ESP intake
2,500 m	82 °C	248.9 m³/h	20 stages, 0.20 m tubing, 90 Hz	0.915 MW	35.2 bar
2,500 m (240 m³/h target)	82 °C	246.3 m³/h	30 stages, 0.20 m tubing, 70 Hz	0.653 MW	35.7 bar
3,400 m	**109 °C**	**235.3 m³/h**	**30 stages, 0.20 m tubing, 60 Hz**	**0.428 MW**	**36.1 bar**

At 2,500 m the brine reaches the plant at 82 °C, so only 35 K of the 40 K cooling cap is usable (injection limit 47 °C). That makes 2 shallow doublets too small for the 30 MW winter peak without backup.

3.2 Design comparison

All designs meet 100% of the heating and cooling demand on all 365 days, with no violations (Table 4, Figure 2). The reference is our assumption: all heat from a gas boiler and all cooling from an electric chiller, 25,943 t CO_2 per year.

Table 4. Annual results of the main designs.

Design	Backup boiler	Grid (GWh/yr)	CO_2 (t/yr)	vs. reference
E2 (recommended): 3 × 3,400 m, direct, gas engine, electric chiller	**0.0%**	**3.9**	**5,533**	**−79%**
F: as E2 with absorption chiller, gas reinjected	0.1%	5.3	1,598	−94%
E: as E2 with absorption chiller	0.0%	0.0	5,706	−78%
Agent's own pick (v5): 3 × 2,500 m, preheat + heat pump, electric chiller	0.0%	11.6	8,184	−68.5%
B: 3 × 2,500 m, preheat + heat pump, absorption chiller	0.0%	12.3	11,068	−57%
A: 2 × 2,500 m, preheat + heat pump	11.0%	13.7	12,759	−51%
D: 3 × 2,500 m, heat pump only	0.0%	36.2	16,706	−36%

3.3 Recommended design

We recommend 3 doublets at 3,400 m with direct heating, a gas engine and an electric chiller: 0% backup boiler and 5,533 t CO_2 per year, 79% below the reference. Reinjecting the gas instead would cut CO_2 by 94%. Table 5 gives the reasons for each decision.

Table 5. Design decisions in E2.

Decision	Choice	Reason
Depth	3,400 m	114 °C reservoir, 109 °C at the plant, so the network is heated straight to 100 °C with no heat pump
Doublets	3	2 deep doublets need about 10% backup boiler; the third covers the 30 MW peak and the outages
Heating	Direct	At 2,500 m the 82 °C brine needs a heat pump top-up of 7–9 GWh of electricity per year
Cooling	Electric chiller	An absorption chiller needs more brine flow, which brings up more gas and CO_2 (5,706 vs. 5,533 t), unless the gas is reinjected
Gas	Gas engine	Burning the methane covers the pump and chiller electricity (grid 3.9 GWh). Associated gas is the largest CO_2 source in every geothermal design
Maintenance	Days 80, 160, 240	Outages start outside the heating peak, never two at once

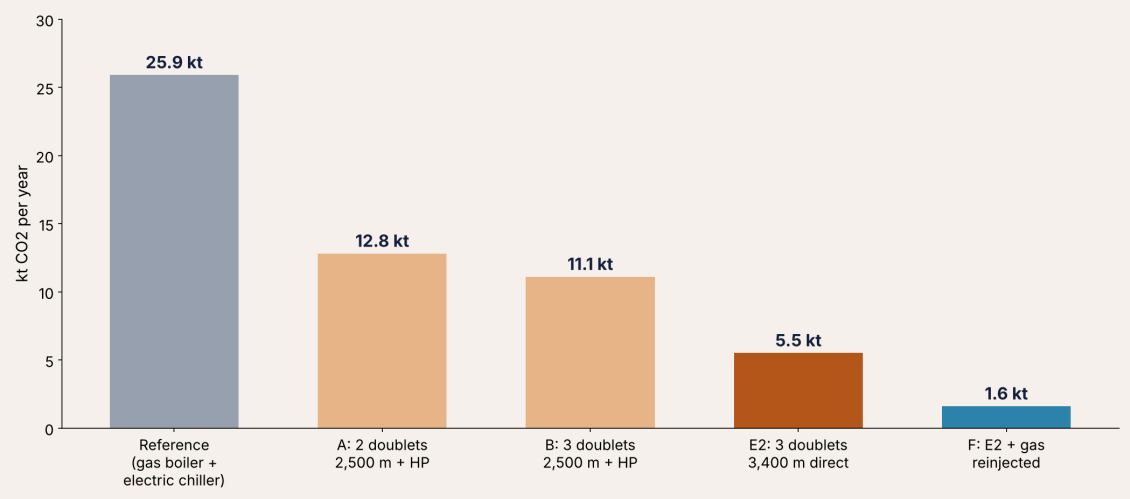


Figure 2. Annual CO_2 emissions of the reference and the main designs.

3.4 How the agent evolved (v1 to v6)

Prompt rules alone did not make the agent reliable; a reviewer step in code did. Each failure below was found in a transcript and fixed in the next version (Appendix A, Table A2). The main run (v5) used 20 LLM calls and 86,959 tokens in 6.4 minutes. The agent's own pick was acceptable but not the best: it never re-evaluated the deep 3-doublet design with outages outside winter. A second, independent run (misghana_run) did find a deep design by itself (3 × 3,400 m, 0.5% backup boiler, 5,747 t CO_2), but its final text quoted two numbers from other designs; all numbers in this report were checked against the tool results.

3.5 Validation

All 12 tests in `tests/test_tools.py` pass on the final code (Table 6). GEMINI and the independent hand calculations differ by 7–50%, and each gap has a known cause (Appendix A, Table A3). The gas engine efficiency is the one real model choice: our simulation uses the hand value (40%), so the GEMINI CHP would give about 1/3 less engine electricity.

Table 6. Automated tests.

Test group	What it checks	Known answer
Challenge document (2)	Demand files: 365 days, peaks, annual totals	30 MW / 111.67 GWh; 10 MW / 23.74 GWh
Hand calculations (6)	Gradient, doublet heat, heat-pump balance, absorption-chiller flow, doublet pressure, gas engine	87 and 114 °C; 13.7 MW; 30 = 20 + 10 MW; ~735 m^3/h; ~52 bar
Limit checks (4)	Flow above 300 or below 50 m^3/h, brine cooling above 40 K, low permeability	A warning, never a silent cap

4 Discussion

4.1 The role of associated gas

The methane in the brine is the largest CO_2 source in every geothermal design. Burning it in a gas engine covers the plant's own electricity, but also releases the 20% native CO_2. Reinjection (design F) cuts emissions to −94%, but the compressor power is not yet modelled, so this result is optimistic.

4.2 GEMINI issues found

We found five GEMINI issues while building the wrapper and reported them in pull request #59 [4]:

1. ESP intake pressure is floored at 0 bar without a warning (`nodal_solver.py:67`, `well_flow.py:295`), which allows non-physical results such as 491 m^3/h at 0 bar intake.
2. `HeatpumpBasic` returns non-physical results (about 390 °C supply, COP 1.0–1.1, or a negative COP) when its source-temperature limit is active.
3. The heat pump and CO_2 corrosion models are missing from the simulator registry, so project JSON files cannot use them.
4. The example gas properties do not match the challenge gas (80% CH_4 / 20% CO_2).
5. Injector wellhead pressure sometimes sits at the solver's lower bound (0.001 bar), so injection pump power shows as 0 MW. Not confirmed as a bug.

4.3 Limitations of our work

- Compressor power for gas reinjection is not modelled, so design F is optimistic.
- The 35 bar bubble point, the productivity index of 14.4 $m^3/h/bar$ and the 1,500 m well spacing are our estimates.
- Daily time step; pump power scales with flow squared between GEMINI runs.
- The acceptance rule (backup boiler under 5%) can favour higher CO_2: with at most 2 doublets the agent chose heat pump only (14,190 t) over 2 deep direct doublets (7,509 t, 10.5% boiler).
- The reviewer checks coverage and acceptance, not optimality: the agent missed design E2.
- In the missing-input test the agent asked a weak question and ended without a final answer.
- The agent's final texts contained wording errors (e.g. a mislabelled cross-check row); all numbers in this report were checked against the tool results.

5 Conclusions

- Three doublets at 3,400 m with direct heating, an electric chiller and a gas engine meet all demand with no backup boiler and 5,533 t CO_2/yr, 79% below the reference.
- Depth is the key geological decision: 3,400 m removes the need for a heat pump.
- The ESP and free gas, not the reservoir, limit each doublet to about 235–249 m^3/h.
- An LLM agent becomes reliable through tools, guard rails and a reviewer in code, not through prompt rules; its output must still be checked against the tools.
- Next step: model reinjection compressor power to confirm the −94% case.

Author contributions. Misghana: subsurface analysis, the hand-calculation tools in `tools.py` (used by the agent's cross-check and the 365-day simulation), the pressure-limit analysis, the 12 automated tests, an independent agent run, and this report. Tareq: the agent (tool loop, reviewer, memory and token handling), the GEMINI wrapper and well sizing, the 365-day simulation, the agent runs v1–v6, and pull request #59.

References

[1] SPE Europe Energy GeoHackathon 2026. GEMINI Hackathon Challenge Requirements and 2027 demand profiles.

[2] GEMINI geothermal digital twin. github.com/GEMINI-Digital-Twin/gemini-geothermal-suite-open

[3] OpenAI. gpt-oss-120b and gpt-oss-20b open-weight models, 2025.

[4] Team GeoBalance. Branch geobalance-agent, pull request #59. https://github.com/GEMINI-Digital-Twin/gemini-geothermal-suite-open/pull/59

Appendix A. Agent evidence and cross-check

All agent sessions used about 402,000 tokens in total on Groq's free tier (about 296,000 in the team runs plus 106,057 in misghana_run); short tests used gpt-oss-20b when the 120b daily limit was reached. Full transcripts are in `hackathon_geobalance/transcripts/`.

Table A1. Agent transcripts.

Transcript	Model	LLM calls	Tokens	What it shows
v5_main_design.md	gpt-oss-120b	20	86,959	Main run: 13 designs; self-correction of the backup boiler from 16.3% to 0.0%; cross-check
main_design.md (v1)	gpt-oss-120b	9	28,122	Outages in midwinter, about 30% boiler, only 3 designs
v2_main_design.md	gpt-oss-120b	9	26,945	Prompt rules alone ignored (23.8% boiler accepted)
v3_with_reviewer.md	gpt-oss-120b	11	36,729	Run made before the reviewer file was installed: same failure as v2
v4_main_design.md	gpt-oss-120b	11	39,252	Stopped by an API-rejected tool call (fixed in v5)
test_offtopic.md	gpt-oss-120b	3	7,712	Bug: the reviewer forced a design answer to an off-topic question
test_offtopic_v6.md	gpt-oss-20b	1	1,720	Pass: declined politely, no tools called
test_missing_input_v6.md	gpt-oss-20b	6	20,174	Partial: took the depth from the inputs, asked one question, no final answer
test_new_constraint_v6.md	gpt-oss-20b	9	40,297	Pass: kept to at most 2 doublets; the reviewer sent it back twice
test_missing_input.md, test_new_constraint.md	gpt-oss-120b	2+1	7,945	Stopped by the daily token limit
misghana_run.md	gpt-oss-120b	20	106,057	Independent run: found a deep 3-doublet design itself (5,747 t CO_2, 0.5% boiler); its final text misquoted two numbers

Example of self-correction (v5): the first design was rejected with "NOT ACCEPTABLE: 0 violation(s), backup boiler 16.3% (limit 5%)" and a warning that one outage fell in the heating season; the agent then moved the outages and added a doublet until the design was accepted.

Table A2. How the agent evolved.

Version	What happened	What we changed
v1: agent + tools	Took both doublets offline from day 0 (midwinter), accepted a design with about 30% backup boiler, tried only 3 designs, misread a GEMINI solver error as a 390 °C temperature	Violation for overlapping outages, warning for heating-season outages, clearer warning text
v2: stricter prompt rules	Ignored the new rules: again 3 designs, 23.8% backup boiler accepted	Reviewer step in code; each result labelled ACCEPTABLE or NOT ACCEPTABLE; exact causes in the cross-check
v3 and v4: reviewer	Explored 8 or more designs, then stopped on Groq's 8,000 tokens-per-minute limit and on an API-rejected tool call	Context trimming, wait-and-retry, rejected calls fed back to the agent, transcript always saved
v5: final	Completed: 13 designs; moved outages out of winter and added a third doublet, cutting the backup boiler from 16.3% to 9.3%, 4.6% and 0.0%	(none: used as the main evidence run)
v6: reviewer scope	First off-topic test: the reviewer forced a design answer to a pizza question	Reviewer only checks design work done in the same session, and drops coverage rules when the user sets constraints

Table A3. Cross-check: GEMINI vs. hand calculation. Case: one doublet at 2,500 m, 246.3 m³/h (agent run v5, step 11).

Quantity	GEMINI	Hand calc.	Diff.	Cause
Producer drawdown	17.1 bar	21.2 bar	+24%	GEMINI: radial inflow with productivity index 14.4 m³/h/bar. Hand: two-well doublet formula (Eq. 3), 1,500 m spacing, half the total pressure difference
Pump power (ESP + injector)	0.66 MW	0.61 MW	−7%	GEMINI: ESP curve at the solved operating point. Hand: pressure × flow / 70%, with an assumed 10 bar wellhead and 5 bar friction
Heat per doublet	9.87 MW	11.28 MW	+14%	Hand uses the full 40 K; in GEMINI the brine reaches the plant at 82 °C, so only 35 K is usable
Gas engine electricity	0.42 MW	0.63 MW	+50%	GEMINI CHP: 80% of fuel energy × 1/3 electric = 26.7% efficiency. Hand: 40% (a modern gas engine)